
Decimal Place Value and Operations — Grade 5 Answer Key

Page 1

1. Four and three hundred eight thousandths; $4 + 0.3 + 0.008$. 2. 7.045; $7 + 0.04 + 0.005$.
3. 0.06, six hundredths.

Page 2

1. The 4 moves from tenths to ones: 0.4 to 4, making its value 10 times as large.
2. 62.5, 625, 6,250. Multiplying by 10, 100, and 1,000 moves each digit one, two, and three places left, respectively. Each digit's value becomes 10, 100, or 1,000 times as large, respectively.
3. 8.24; digits shift one place right in value, so each digit's value is one tenth as much.

Page 3

1. =. 2. $<$ ($5.206 < 5.260$). 3. 2.005, 2.05, 2.15, 2.5.

Page 4

1. Circle the 7 in the tenths place. The next digit is 4, so round to 8.7.
2. Circle the 0 in the hundredths place. The next digit is 3, so round to 19.50.
3. For the nearest whole mile, circle the 6; the tenths digit is 9, so the answer is 7 miles. For the nearest tenth, circle the 9; the hundredths digit is 5, so rounding up gives 7.0 miles.

Page 5

1. 17.287. 2. 20.552. 3. 4.375 miles.

Page 6

1. 12.154. 2. 5.525. 3. 11.775 liters.

Page 7

Estimates may vary; these are examples using nearby whole numbers.

1. Estimate: $45 + 9 = 54$. Exact: $45.670 + 8.905 = 54.575$.
2. Estimate: $30 - 15 = 15$. Exact: $30.200 - 14.786 = 15.414$.
3. Estimate: $10 - 6 = 4$. Subtract the known addend: $10.025 - 6.400 = 3.625$.

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1. $(3 + 0.2 + 0.06) \times 4 = 12 + 0.8 + 0.24 = 13.04$.
2. $(10 + 2 + 0.4 + 0.005) \times 6 = 60 + 12 + 2.4 + 0.03 = 74.430$.
3. $(2 + 0.7 + 0.05) \times 8 = 16 + 5.60 + 0.40 = 22.00$, so the cost is \$22.00.

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1. 48.2. Each digit moves two places left, so each digit's value becomes 100 times as large.
2. 7,090. Each digit moves three places left, so each digit's value becomes 1,000 times as large.
3. 100. In changing 5.6 to 560, the 5 moves from ones to hundreds and the 6 from tenths to tens; each digit's value becomes 100 times as large.

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1. $8.4 \div 4 = 2.1$. Check: $2.1 \times 4 = 8.4$.
2. $36.75 \div 5 = 7.35$. Check: $7.35 \times 5 = 36.75$.
3. $14.4 \div 6 = 2.4$ pounds per bag. Check: $2.4 \times 6 = 14.4$ pounds.

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1. 6.35; each digit moves one place right, making its value one tenth as much.
2. 0.0408; each digit moves two places right, making its value one hundredth as much.
3. 0.725; each digit moves three places right, making its value one thousandth as much.

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Estimates may vary; these are examples.

1. Estimate: $5 \times 2.5 = 12.5$. Since $48 \times 25 = 1,200$ and both factors represent tenths, the product is 1,200 hundredths, or $12.00 = 12$.
2. Estimate: $0.4 \times 0.7 = 0.28$. Since $36 \times 7 = 252$ and hundredths times tenths gives thousandths, the product is 252 thousandths, or 0.252.
3. Estimate: $4 \times 1 = 4$ square feet. Area is length times width. Since $35 \times 12 = 420$ and both factors represent tenths, $3.5 \times 1.2 = 4.20$, or 4.2 square feet.

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1. Multiply both numbers by 10 to form an equivalent division: $5.6 \div 0.7 = 56 \div 7 = 8$.
2. $12.48 \div 0.6 = 124.8 \div 6$. Since $120 \div 6 = 20$ and $4.8 \div 6 = 0.8$, the quotient is 20.8.
3. $7.5 \div 0.5 = 75 \div 5 = 15$ pieces. Multiplying both numbers by 10 preserves the quotient.

Page 14

1. \$24.84. 2. \$6.32. 3. \$6.90.

Page 15

1. 15.15 feet. 2. 1.625 liters. 3. 25.5 miles.

Page 16

1. Subtract the difference from the starting number: $15.200 - 9.875 = 5.325$.
2. Subtract the known addend from the sum: $8.005 - 2.670 = 5.335$.
3. Divide the product by the known factor: $7.2 \div 4 = 1.8$.

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1. $3(4.25) + 2.50 = 15.25$, so \$15.25.
2. $12.5 - 7.85 = 4.65$ miles.
3. $9.6 \div 6 = 1.6$ liters.

Page 18

1. False: $0.45 < 0.5$ because $0.45 < 0.50$; 45 hundredths is less than 50 hundredths.
2. False: $2.4 \times 0.2 = 0.48$, smaller than 2.4. Multiplying by 0.2 takes two tenths of the original positive number.
3. 9.74 is reasonable because $48.7 \div 5$ is close to $50 \div 5 = 10$, whereas 0.974 is less than 1. Exactly, $48.7 \div 5 = 9.74$; check: $9.74 \times 5 = 48.7$.

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1. $6.249 \rightarrow 6.25$; $6.25 + 3.75 = 10$.
2. $9.6 \times (2 + 0.7 + 0.05) = 19.2 + 6.72 + 0.48 = 26.4$.
3. $24.8 \div 8 = 3.1$ meters per piece; check: $3.1 \times 8 = 24.8$ meters.

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1. Area is length times width: $6.25 \times 3.2 = 6.25 \times (3 + 0.2) = 18.75 + 1.25 = 20$ square meters.

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2. The notebooks cost $3 \times 4.85 = 14.55$ dollars. Including the pen, the total is $14.55 + 2.75 = 17.30$ dollars. Subtract this from the starting money: $50.00 - 17.30 = 32.70$, so \$32.70 remains.
 3. Example: 4.065; it is greater than 4.060 and less than 4.070; four and sixty-five thousandths.